

WEEK 3 (Praveen 2520030406)

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        printf("fork() failed\n");
        exit(1);
    }

    else if (pid == 0)
    {
        // Child Process
        printf("\n--- Child Process ---\n");
        printf("Child PID: %d\n", getpid());
        printf("Child PPID: %d\n", getppid());

        printf("Child is running...\n");

        sleep(5);

        printf("Child is terminating...\n");
        exit(0);
    }

    else
    {
        // Parent Process
        printf("\n--- Parent Process ---\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        printf("Parent is waiting for child...\n");

        wait(NULL);

        printf("Child terminated.\n");
        printf("Parent process terminating...\n");
    }

    return 0;
}
```

Output:

```
[praveenbusala@Praveens-MacBook-Air-2 CLG % nano week3.c
[praveenbusala@Praveens-MacBook-Air-2 CLG % gcc week3.c
[praveenbusala@Praveens-MacBook-Air-2 CLG % ./a.out
Parent Process Started
Parent PID: 24094

--- Parent Process ---
Parent PID: 24094
Child PID: 24095
Parent is waiting for child...

--- Child Process ---
Child PID: 24095
Child PPID: 24094
Child is running...
Child is terminating...
Child terminated.
Parent process terminating...
praveenbusala@Praveens-MacBook-Air-2 CLG %
```